

HENRY JIANG

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Education

Georgia Institute of Technology

Atlanta, GA

- B.S. in Computer Science | Concentrations: Artificial Intelligence, Computational Media May 2026
- GPA: 3.66 | Graduating with Highest Honors, Dean's List 2023, 2024, 2025, Eta Kappa Nu Scholar, IEEE Member

Skills

Languages: Python, Java, JavaScript, Typescript, C, C#, C++, HTML, CSS, R, SQL, Bash

Tools: Git, JUnit, Jira, Docker, Splunk, Firebase, Tensorflow, Pytorch, Scikit-learn, SciPy, Matplotlib, Seaborn, Pytorch Geometric, Pandas, Numpy, Splunk, Selenium, OpenCV, ASE, MATLAB

Frameworks: React, Next.js, Three.js, Node.js, REST APIs, Angular, Android Studio, Figma, Firebase, PostgreSQL, MySQL, MongoDB

Experience

GT Fung Research Group

Aug 2023 – May 2026

Undergraduate Research Assistant

Atlanta, GA

- Developed 2 graph neural network (GNN) packages and publishing paper for crystal structure prediction achieving 90%+ accuracy on materials datasets with 10,000+ atomic configurations using PyTorch Geometric, ASE, and Pyswarms
- Solved inverse design bottleneck for unknown lattice parameters with a hybrid particle swarm optimization algorithm, achieving 80% structure match rate and reducing prediction error by 50% compared to baseline methods
- Engineered data preprocessing pipeline processing 500+ crystal structures from Materials Project database, discovering 10+ novel ground-state configurations through automated DFT validation

Blender Foundation

Apr 2024 – Dec 2025

Extension Developer & Contributor

- Designed and implemented 2 Blender addons using Python and Blender API, enhancing modeling and animation workflow efficiency by 20%+ for 1,000+ users by automating IK/FK snapping and smart topology based selection tools
- Collaborated with senior developers via Github and Gitea to report and resolve 3 high-priority animation module bugs and help optimize weight painting scripts issues, reducing pipeline open issues by 5%

NASA

Jun 2024 – Aug 2024

Information Systems Security Intern

Kennedy Space Center, FL

- Engineered 4 production Splunk dashboards for Launch Control Center monitoring 100,000+ security events, implementing real-time alert systems with custom SPL queries and automated data aggregation pipelines
- Strengthened cybersecurity compliance by completing SSP and SCP revisions for 3 NE-XC sector teams with cross-collaboration, analyzing NIST/FISMA requirements and drafting 20+ PO&AMs for vulnerabilities
- Built automated data preprocessing pipeline for security event classification, achieving 75% reduction in false positive alerts through threshold optimization and time-series analysis techniques

Projects

DARWIN: Agentic Evolutionary GPT | OpenAI API, LLMs, Genetic Algorithms | [Github](#) | [arXiv](#)

- Designed agentic evolutionary AI with self-improving agents optimizing NN architecture training code and hyperparameters achieving 1.2%+ MFU improvement and 2%+ perplexity improvement on fully trained models
- Engineered agentic CI/CD style debugging pipeline, persistent JSON memory system, and bidirectional communication interface for dynamic upgrade requests from agents (datasets, packages, scripts)

Mandelbrotian Volatility Options Trading Model | QuantConnect, Python, Scipy, Scikit-learn | [Github](#) | [Quantconnect](#)

- Engineered a fractal regime-detection OTM options strategy with Hurst exponents (DFA), Hill tail-index, and multifractal spectral width to classify volatility regimes achieving 55%+ CAGR and 1.18 Sharpe across 105 trades over 5 years ('20-'25)
- Built an XGBoost-based trade filter trained on 27 technical features with scikit-learn cross-validation, regime-adaptive stop-losses and dynamic position sizing delivering +36% average trade PnL and 1.45+ Sortino across MAG7 tech equities

Viper: Artist Social Media with Image Poisoning | Postgres, React, MongoDB, Next.js, Node.js, AWS, Figma | [wip](#)

- Developed a full-featured social ecosystem with React, Next.js, and Node.js including real-time messaging, notification, and profile management, maintaining strict "No-LLM" development integrity and server-side image poisoning pipelines.
- Deployed a secure, cloud-native backend on AWS with PostgreSQL and MongoDB, managing complex state transitions and persistent user sessions across 10+ core feature modules.